
SELECTED PUBLICATIONS (# Equal Contribution, * Corresponding Author)

2026

1. Muhammad Sarfraz, Danish Ali, Muhammad Ayyaz, **Sabir Ali Siddique** “Indirubin Derivatives: Synthesis, Physicochemical Profiling, Toxicity Assessment, and Molecular Docking Investigations Against EGFR and HIV Target Substrates”, **ChemistrySelect** (2026).
2. Shaaban K. Mohamed*, **Sabir Ali Siddique***, Muhammad Sarfraz, Etify A. Bakhite, Suzan Abuelhassan, Islam S. Marae, Abdelhamid A. E. Soliman, Esraa Khamies, Aseel Smerat, Basma A. A. Balboul, Joel T. Mague, Youness El Bakri* “Synthesis, Structural, Electronic, and Medicinal Characterization of a Thiophene-Bearing Polycyclic Compound: An Approach to Discover Potential Lead Compounds for Precision Cancer Therapy”, **ChemistrySelect** (2026).
3. **Sabir Ali Siddique#**, Shanza Fatima; Ejaz Ahmed; Rahman Shah Zaib Saleem; Muhammad Bilal Ahmed Siddique*, Muhammad Arshad*, “Quantum chemical engineering of enhanced nonlinear optical responses in alkali metal-doped diazadioxacirculenes for molecular photonics”, **Journal of Molecular Graphics and Modelling** (2026).

2025

4. **Sabir Ali Siddique#**, Kashaf Afzal, Ali Rauf, Muhammad Bilal Ahmed Siddique*, Nida Ambreen, Muhammad Arshad*, “Coinage Metal-Functionalized P-Flowers with Remarkable Hyperpolarizability for Advanced Photonic Applications: A DFT Study”, **PCCP (CP-ART-10-2025-002841)**, In peer review (25-04-2026).
5. Asad Ullah, Muhammad Arif Ali, **Sabir Ali Siddique**, Muhammad Ibrahim, Hong Liang Xu, Abdul Rauf, Amal M. Al-Mohaimeed, Wedad A. Al-onazi, Muhammad Arshad, “Quantum Chemical Insights into Metal-Ion Enhanced NLO Response of a Fluorescent Probe for Advanced Sensing Application.” (includes excited-state property analysis via TD-DFT), **Journal of Fluorescence** (2025), <https://doi.org/10.1007/s10895-025-04382-z>
6. Surayyo R. Razzaqova, **Sabir Ali Siddique**, Jamoliddin N. Todjiyev, Shakhnoza A. Kadirova, Dilnoza S. Rakhmonova, Aziz B. Ibragimov, Adkhamjon S. Normamatov, Abror Kh. Ruzmetov, Yuldosh Yu. Yakubov, Bakhtiyar T. Ibragimov, C. Balakrishnan, Hatem A. Abuelizz, Rashad Al-Salahi, Junkuo Gao, Youness El Bakri, “Synthesis, X-ray Structure Investigation, Hirshfeld Surface Analysis and Theoretical Calculations of Metal Complexes on the Base of 2-Aminobenzoxazole” **ChemistrySelect** (2025), <https://doi.org/10.1002/slct.202500087>

7. **Sabir Ali Siddique**, Shaaban K Mohamed, Muhammad Sarfraz, Etify A Bakhite, Islam S Marae, Abdelhamid AE Soliman, Esraa Khamies, Ahmed F Selim, Maha QM Qahtan, Hatem A Abuelizz, Rashad Al-Salahi, Joel T Mague, Youness El Bakri, "Synthesis, investigation of the crystal structure, DFT and in silico medicinal potential of nicotinonitrile substituted quinazolidione as potential anticancer scaffold." **Molecular Physics** (2025), <https://doi.org/10.1080/00268976.2024.2448566>
8. Adnan, Muhammad, Iqra Munir, Riaz Hussain, Zobia Irshad, Riaz Hussain, **Sabir Ali Siddique**, and Hany W. Darwish, "Role of Near-IR Sensitive Asymmetric A-DA' D- π -A-type Non-Fullerene Acceptors for Efficient Organic Solar Cells: A Theoretical Insight." (includes excited-state property analysis via TD-DFT), **Journal of Computational Biophysics and Chemistry** (2025), <https://doi.org/10.1142/S2737416524500546>

2024

9. **Sabir Ali Siddique**, Muhammad Bilal Ahmed Siddique, Ejaz Ahmed, Asad Ullah, Ali Rauf, Muhammad Arif Ali, Tariq Mahmood, Abdul Rauf and Muhammad Arshad, Quantum Chemical Investigation for Enhanced Electrochemical Sensing of Toxic Gases by HexaazaphenH₂, (includes excited-state property analysis via TD-DFT), **New Journal of Chemistry** (2024), <https://doi.org/10.1039/D4NJ02845G>
10. Wajiha Akbar, Shahana Ehsan, **Sabir Ali Siddique***, Muhammad Sarfraz, Faiqa Shaheen, Ayesha Shafqat, Shahnaz*, Muhammad Bilal Ahmed Siddique, Ayesha Saeed, Rashad Al-Salahi, Youness El Bakri, Solid Phase Synthesis, DFT Calculations, Molecular Docking, and Biological Studies of Symmetrical N₂,N₄,N₆-Trisubstituted-1,3,5-triazines, **ACS Omega** (2024), <https://doi.org/10.1021/acsomega.4c01980>
11. Youness El Bakri*, **Sabir Ali Siddique***, Shaaban K. Mohamed, Muhammad Sarfraz, Hatem A. Abuelizz, Rashad Al-Salahi, Joel T. Mague and Eman A. Ahmed, Synthesis, investigation of the crystal structure, DFT calculations, and in silico medicinal potential of hydrazono- and aminomethylene substituted pyrazolidine-3,5-diones as potential anticancer scaffolds, **New Journal of Chemistry** (2024), <https://doi.org/10.1039/D4NJ01839G>
12. Muhammad Sarfraz, Muhammad Ayyaz, Abdul Rauf, Asma Yaqoob, Tooba, Muhammad Arif Ali, **Sabir Ali Siddique**, Ashfaq Mahmood Qureshi, Muhammad Hassan Sarfraz, Reem M. Aljowaie, Saeedah Musaed Almutairi, and Muhammad Arshad, New Pyrimidinone Bearing Aminomethylenes and Schiff Bases as Potent Antioxidant, Antibacterial, SARS-CoV-2, and COVID-19 Main Protease MPro Inhibitors: Design, Synthesis, Bioactivities, and Computational Studies, **ACS Omega** (2024), <https://doi.org/10.1021/acsomega.3c09393>
13. Muhammad Ayyaz, Muhammad Sarfraz, Muhammad Arshad, Asma Yaqoob, **Sabir Ali Siddique**, Safdar Hussain, Muhammad Arif Ali, Ashfaq Mahmood Qureshi, Abdul Rauf, "Design, synthesis, in-vitro biological

- screening, and in-silico studies of 2-thioxodihydropyrimidinone based new aminomethylene scaffolds, **Journal of Molecular Structure** (2024), <https://doi.org/10.1016/j.molstruc.2023.137153>
14. Shaaban K. Mohamed*, **Sabir Ali Siddique***, Subramani Karthikeyan, Eman A. Ahmed, Omran A. Omran, Joel T. Mague, Rashad Al-Salahi, Youness El Bakri, Synthesis, X-ray crystallography, computational investigation on quinoxaline derivatives as potent against adenosine receptor A2AAR, **Journal of Biomolecular Structure and Dynamics** (2024), <https://doi.org/10.1080/07391102.2024.2314268>
 15. **Sabir Ali Siddique**, Babar Ali, Muhammad Bilal Ahmed Siddique, Ali Rauf, Riaz Hussain, Muhmmad Arif Ali, Tariq Mahmood, Sadia Altaf, Abdul Rauf, Seham J.F. Alanazi, Amal M. Al-Mohaimeed, Xin Liu, Muhmmad Arshad, Discovery of pyrrole-triphenylamine based novel organic sensitizers for dye-sensitized solar cells: A first principal study, (includes excited-state property analysis via TD-DFT), **Materials Science in Semiconductor Processing** (2024), <https://doi.org/10.1016/j.mssp.2024.108173>
 16. Galiya R. Tajieva, **Sabir Ali Siddique***, Aziz B. Ibragimov, Shaaban K. Mohamed, Mavlyuda R. Ibragimova, Hatem A. Abuelizz, Rashad Al-Salahi, Jamshid M. Ashurov, Bakhtiyar T. Ibragimov, Junkuo Gao, Youness El Bakri, Synthesis and characterization of novel quinaldic acid based Ni, Fe, and Cr Complexes: A computational and experimental study, **Inorganic Chemistry Communications** (2024), <https://doi.org/10.1016/j.inoche.2024.112590>

2023

17. Humera Baig, Rimsha Irfan, Alvina Rasool, Syed Zajif Hussain, **Sabir Ali Siddique**, Javed Iqbal, Meshari Alazmi, Nawaf Alshammari, Amira Alazmi, Amer AlGhadhban, Abdel Moneim E. Sulieman, Kamaleldin B. Said, Habib-ur-Rehman, Rahman Shah Zaib Saleem, Synthesis, photophysical, voltammetric, and DFT studies of 4-aminochalones, (includes excited-state property analysis via TD-DFT), **Journal of Photochemistry and Photobiology A: Chemistry** (2023), <https://doi.org/10.1016/j.jphotochem.2023.114790>
18. Youness El Bakri, Malahat Kurbanova, **Sabir Ali Siddique**, Subramani Karthikeyan, Abel Maharramov, Nacaf Ramazanzade, Synthesis, virtual screening, and computational approach of 6-(4-methoxyphenyl)-4-phenyl-1,2-dihydropyrimidin-2-one as a potential target for thioredoxin glutathione reductase (TGR), **Journal of Molecular Structure** (2023), <https://doi.org/10.1016/j.molstruc.2023.135623>
19. Babar Ali[#], **Sabir Ali Siddique[#]** ([#] equal contribution), Muhammad Bilal Ahmed Siddique, Sami Ullah, Muhammad Arif Ali, Abdul Rauf, Muhammad Arshad Kamran, Muhammad Arshad, Insight on the structural, electronic, and optical properties of Zn, Ga-doped/dual-doped graphitic carbon nitride for visible-light applications, (includes excited-state property analysis via TD-DFT), **Journal of Molecular Graphics and Modelling** (2023), <https://doi.org/10.1016/j.jmgm.2023.108603>

20. Adnan Muhammad, Fatima Qandeel, Irshad Zobia, Hussain Riaz, Yaqoob Junaid, Khan Muhammad Usman, Ansari Nooruddin, **Sabir Ali Siddique**, Lim Jongchul, Role of Aromatic Heterocyclic Core- Based Materials as Donors for Organic and as Hole-Transporting Materials for Perovskites Solar cells, **Journal of Computational Biophysics and Chemistry** (2023), <https://doi.org/10.1142/S2737416523420012>
21. Muhammad Adnan, Muhammad Faisal, Riaz Hussain, Muhammad Usman Khan, Junaid Yaqoob, Fakhar Hussain, **Sabir Ali Siddique**, Umar Farooq, Churl-Hee Cho, Jongchul Lim, Modeling of Conjugated Core-Based Non-Fullerene Acceptor Materials for Organic Photovoltaic Applications, **Journal of Computational Biophysics and Chemistry** (2023), <https://doi.org/10.1142/S2737416523500370>

2022

22. **Sabir Ali Siddique**, Sadia Altaf, Ejaz Ahmed, Muhammad Bilal Ahmed Siddique, Riaz Hussain, Xin Liu, Abdul Rauf, Muhammad Arshad, Discovery of versatile bat-shaped acceptor materials for high-performance organic solar cells a DFT approach, (includes excited-state property analysis via TD-DFT), **International Journal of Energy Research** (2022), <https://doi.org/10.1002/er.8049>
23. Youness El Bakri, Malahat Musrat Kurbanova, **Sabir Ali Siddique**, Sajjad Ahmad, Souraya Goumri-Said, One-pot synthesis, X-ray crystal structure, and identification of potential molecules against COVID-19 main protease through structure-guided modeling and simulation approach, **Arabian Journal of Chemistry** (2022), <https://doi.org/10.1016/j.arabjc.2022.104230>
24. **Sabir Ali Siddique**, Muhammad Arshad, Muhammad Yasir Mehboob, Muhammad Adnan, Riaz Hussain, Babar Ali, Muhammad Bilal Ahmed Siddique, Xin Liu, Efficient tuning of zinc phthalocyanine-based dyes for dye-sensitized solar cells: a detailed DFT study, (includes excited-state property analysis via TD-DFT), **RSC Advances** (2022), <https://doi.org/10.1039/D1RA04529F>
25. Faheem Abbas, Usman Ali, Aqsa Tallat, Hafiz Muhammad Rizwan Ahmad, **Sabir Ali Siddique**, Zonish Zeb, Muhammad Bilal Ahmed Siddique, An optoelectronic study to design better benzodithiophene (BDT) donor unit based non-fullerene organic solar cells (OSCs): the DFT approaches, **Chemical Papers** (2022), <https://doi.org/10.1007/s11696-022-02224-x>
26. Hasnain Sajid[#], **Sabir Ali Siddique**[#] (**# equal contribution**), Ejaz Ahmed, Muhammad Arshad, Mazhar Amjad Gilani, Abdul Rauf, Muhammad Imran, Tariq Mahmood, DFT outcome for comparative analysis of Be₁₂O₁₂, Mg₁₂O₁₂, and Ca₁₂O₁₂ nanocages toward sensing of N₂O, NO₂, NO, H₂S, SO₂, and SO₃ gases, **Computational and Theoretical Chemistry** (2022), <https://doi.org/10.1016/j.comptc.2022.113694>
27. **Sabir Ali Siddique**, Hasnain Sajid, Mazhar Amjad Gilani, Ejaz Ahmed, Muhammad Arshad, Tariq Mahmood, Sensing of SO₃, SO₂, H₂S, NO₂, and N₂O toxic gases through aza-macrocycle via DFT

calculations, **Computational and Theoretical Chemistry** (2022),
<https://doi.org/10.1016/j.comptc.2022.113606>

2021

28. Maria Akram, **Sabir Ali Siddique***, Javed Iqbal*, Riaz Hussain*, Muhammad Yasir Mehboob, Muhammad Bilal Ahmed Siddique, Babar Ali, Ayesha Hanif, Muhammad Sajid, Sana Shoukat, End-capped engineering of bipolar diketopyrrolopyrrole based small electron acceptor molecules for high performance organic solar cells, **Computational and Theoretical Chemistry** (2021),
<https://doi.org/10.1016/j.comptc.2021.113242>
29. **Sabir Ali Siddique**, Muhammad Usman Alvi, Muhammad Yasir Mehboob, Babar Ali, Abdul Rauf, Muhammad Bilal Ahmed Siddique, Riaz Hussain, Muhammad Arshad, Xin Liu, Deciphering the role of end-capped acceptor units for amplifying the photovoltaic properties of donor materials for high-performance organic solar cell applications **Computational and Theoretical Chemistry** (2021),
<https://doi.org/10.1016/j.comptc.2021.113454>
30. Madiha Batool*, Shazia Khurshid, Zahid Qureshi, Ali Hassan, Muhammad Bilal Ahmed Siddique, **Sabir Ali Siddique***, Study of biogenically fabricated transition metal oxides nanoparticles on oral cavity infectious microbial strains, **Inorganic and Nano-Metal Chemistry** (2021),
<https://doi.org/10.1080/24701556.2020.1811729>
31. Madiha Batool*, Shazia Khurshid, Walid M. Daoush*, **Sabir Ali Siddique***, Tariq Nadeem, Green Synthesis and Biomedical Applications of ZnO Nanoparticles: Role of PEGylated-ZnO Nanoparticles as Doxorubicin Drug Carrier against MDA-MB-231(TNBC) Cells Line, **Crystals** (2021),
<https://doi.org/10.3390/cryst11040344>
32. **Sabir Ali Siddique**, Muhammad Arshad, Sabiha Naveed, Muhammad Yasir Mehboob, Muhammad Adnan, Riaz Hussain, Babar Ali, Muhammad Bilal Ahmed Siddique, Xin Liu, "Efficient tuning of zinc phthalocyanine-based dyes for dye-sensitized solar cells: a detailed DFT study" **RSC Advances** (2021),
<https://doi.org/10.1039/D1RA04529F>

2020

33. **Sabir Ali Siddique**, Muhammad Bilal Ahmed Siddique, Riaz Hussain, Xin Liu, Muhammad Yasir Mehboob, Zobia Irshad, Muhammad Adnan, Efficient tuning of triphenylamine-based donor materials for high-efficiency organic solar cells, **Computational and Theoretical Chemistry** (2020)
<https://doi.org/10.1016/j.comptc.2020.113045>
34. Muhammad Bilal Ahmed Siddique, Riaz Hussain, **Sabir Ali Siddique**, Muhammad Yasir Mehboob, Zobia Irshad, Javed Iqbal, Muhammad Adnan, Designing Triphenylamine-Configured Donor Materials with

Promising Photovoltaic Properties for Highly Efficient Organic Solar Cells, **ChemistrySelect** (2020), <https://doi.org/10.1002/slct.202001989>